AI Appointment System

Tools used and how each one supports the hospital

Lakshmi Multispecialty Hospital · Virudhunagar, Tamil Nadu · Prepared 9 October 2026

Summary

Patients can book a doctor appointment in four ways – a phone call to the AI voice receptionist, the AI chat on the website, the online booking form, or a WhatsApp request from the patient page. Every channel uses the **same booking engine and the same appointment database**, so a time booked in one place disappears everywhere else at once and two patients can never get the same slot.

This document lists every tool in the system, in plain language: what it is, how it helps the hospital, and whether it is working today or still needs setup.

Phone call (AI voice)	Website AI chat	Online booking form	Mobile app	Admin dashboard
All channels → one booking engine (Node.js + Express) → one MySQL appointment database → appointment number LMH-0001, LMH-0002 …				

At a glance	
Doctors	Mr. Vijul (Orthopaedic) · Mr. Vishwa (Diabetologist) · Ms. Nivi (Heart Specialist)
Languages	Tamil, English and Hindi
Appointment numbers	LMH-0001, LMH-0002 … issued only after the database saves the booking
Automated tests	60 tests, all passing on both MySQL and SQLite
Source code	Saved on GitHub (private secrets are never uploaded)

1. Core system (the booking engine)

These run the hospital system itself.

Tool	What it is	How it supports the hospital	Status
Node.js 24	The program that runs the hospital server on the computer.	Runs the booking engine, website and all APIs. One program serves the website, AI chat, voice agent and admin together.	**Working**
Express	Web-server toolkit for Node.js.	Delivers the web pages and answers requests such as "show free slots" or "book this appointment". Adds security headers and rate limits.	**Working**
Booking engine (custom code)	The hospital's own rules written in code.	Checks real free slots, doctor timings, doctor leave, Sunday holiday, today's cut-off and the next working day. Never invents a time; issues LMH numbers.	**Working**
dotenv (.env file)	A private settings file on the computer.	Keeps database password, OmniDimension secrets, WhatsApp token and admin password out of the website and out of GitHub.	**Working**

2. Database

Tool	What it is	How it supports the hospital	Status
MySQL / MariaDB (XAMPP)	The appointment database, run by XAMPP on this computer.	Stores doctors, timings, leave, appointments, notifications and call records. A unique rule on each time slot makes double booking impossible, even if two people book at the same second.	**Working**
XAMPP Control Panel	Windows app that starts and stops MySQL.	Lets staff start the database with one click before using the system.	**Working**
mysql2	Connector between Node.js and MySQL.	Lets the booking engine read and save appointments safely (locked transactions for each booking).	**Working**
SQLite (built into Node.js)	A small file-based database.	Used only for automated tests and quick trials, so testing never touches real patient data.	**Working**

3. Patient-facing channels

Tool	What it is	How it supports the hospital	Status
Website (HTML, CSS, JavaScript)	The hospital website pages.	Home, Doctors, Doctor Schedule (live free slots), Book Appointment, AI Voice Receptionist, Appointment Status and Contact. Mobile-friendly with large buttons.	**Working**
Patient page (index.html)	A simple one-page site for patients.	Big Call and WhatsApp icon buttons (no number shown as text), doctor list from the database, WhatsApp request form, "A+ Larger text" for older patients, emergency 108 box.	**Working**

Tool	What it is	How it supports the hospital	Status
Hind Madurai font (Google Fonts)	A typeface that supports Tamil.	Makes English and Tamil text clear and easy to read on phones.	**Working**
AI Chat Receptionist (built-in)	Floating chat on the website.	Books, checks, reschedules and cancels appointments in Tamil, English or Hindi. Shows only real free slots from the database; no outside AI service or key needed.	**Working**
Expo / React Native (Expo Go)	Tools for building the mobile app.	A phone app with Book, Doctors, AI Chat and Status screens, using the same booking engine. Opens in the free Expo Go app for testing.	**Built – not yet tested on a phone**

4. AI voice receptionist and messaging

Tool	What it is	How it supports the hospital	Status
OmniDimension	AI voice agent platform that answers phone calls.	Answers incoming calls in Tamil, English and Hindi, asks doctor → date → morning/evening → time, checks real free slots through the hospital's voice API, reads back the details and books only after the caller says yes.	**Ready – needs setup**
Voice API (/api/voice/...)	Secure web addresses the AI agent calls during a call.	list_doctors, check_availability, book_appointment, find, reschedule, cancel and staff_callback. Protected by a secret key; tested with simulated calls.	**Working**
Post-call webhook	Address that receives call results after each call.	Saves only the useful details (patient, doctor, date, time, appointment number, language) – never full transcripts – and marks each record "verified" against the database.	**Ready – needs setup**
WhatsApp click-to-chat (wa.me)	WhatsApp link used by the patient page.	Opens WhatsApp with the patient's appointment request already typed. Clearly marked "a request, not a confirmed booking".	**Working**
WhatsApp Cloud API (Meta)	Official WhatsApp sending service.	Will send staff alerts for new booking, confirmation, cancellation, rescheduling, doctor leave, reminders and "staff attention". Every message is logged with its status.	**Ready – needs setup**

5. Hospital staff tools

Tool	What it is	How it supports the hospital	Status
Admin dashboard (/admin)	Password-protected staff screen.	View today's, tomorrow's and upcoming appointments; search and filter by doctor or date; confirm, cancel, reschedule; mark doctor leave; see free slots, AI voice bookings, call records and WhatsApp status.	**Working**
Doctor-leave handling	Part of the admin dashboard.	Blocks bookings on leave days, flags affected patients and suggests each one a different next free slot that staff can apply in one click.	**Working**

Tool	What it is	How it supports the hospital	Status
Reminders	Automatic reminder job.	Sends one reminder per appointment about 24 hours before (via WhatsApp once connected).	**Ready – needs setup**

6. Connection, hosting and safety tools

Tool	What it is	How it supports the hospital	Status
Cloudflare Tunnel (cloudflared)	Creates a temporary public web address for this computer.	Lets OmniDimension and patients' phones reach the system while it runs on the hospital PC. The address changes each time it restarts.	**Working (test link)**
Vercel	Cloud hosting service.	Planned permanent public website address (needs a cloud MySQL database).	**Planned**
Git and GitHub	Version history and online backup of the code.	Every change is saved and can be restored. The .env file and database are never uploaded.	**Working**
Security protections (built-in)	Secret keys, login cookies, rate limits and browser rules.	Voice API requires a secret key; admin login is protected; repeated requests are limited; private phone numbers are not shown as text on the website.	**Working**
Automated tests (Node test runner)	60 checks that run in about 30 seconds.	Prove booking, duplicate prevention, doctor leave, next-day logic, cancellation, rescheduling, Tamil/Hindi requests and security still work after every change.	**Working**
Google Chrome (headless)	Browser used for automatic screen checks.	Confirms pages fit a 360-pixel phone screen with no sideways scrolling and no errors.	**Working**

7. How each hospital requirement is supported

Requirement	Supported by
Never invent an appointment	Booking engine checks MySQL for every slot; voice agent and chat can only offer times the database returns.
No double booking	Unique slot rule in MySQL + locked transactions. Tested with simultaneous bookings: only one succeeds, the other is offered the next real slot.
Today's booking closed → next working day	Booking engine skips past times, the 30-minute cut-off and Sundays, and offers the first real slot on the next working day.
Doctor on leave	Admin dashboard "Mark doctor leave" + booking engine refuses leave days and offers the next free day.
Appointment numbers LMH-0001 …	Issued by the database only after the booking is saved; no gaps.
Tamil, English, Hindi	OmniDimension agent (voice), built-in AI chat, website language switch, Hind Madurai font.
Confirm before booking (voice)	Voice API requires "patient_confirmed = true"; otherwise it only reads back the details.

Requirement	Supported by
Private phone numbers	Numbers kept in .env; website shows icon buttons only, never the number as text.
Staff notifications	Notification service + WhatsApp Cloud API (waiting for Meta credentials); every message logged in the admin dashboard.
Older patients / mobile users	Large buttons, big text, "A+ Larger text" option, layout tested at 360-pixel phone width.

8. What is working and what still needs setup

Item	Status	What is needed
Booking engine, database, website, AI chat, admin	**Working**	Nothing – start XAMPP MySQL and run "npm start".
Patient page with Call and WhatsApp buttons	**Working**	Nothing.
Voice API (tested through the tunnel)	**Working**	Test booking LMH-0001 was created successfully.
OmniDimension agent in the dashboard	**Ready – needs setup**	Add the remaining Custom API tools and post-call webhook, then make the first real test call.
WhatsApp staff notifications	**Ready – needs setup**	Meta WhatsApp Cloud API token, phone-number ID and an approved message template.
Permanent public website	**Planned**	Cloud hosting (e.g. Vercel) with a cloud MySQL database.
Mobile app	**Built – not yet tested on a phone**	Test on a phone with Expo Go; publishing to Play Store / App Store is a later step.

9. Daily start-up (on the hospital computer)

1. Open the **XAMPP Control Panel** and click **Start** next to MySQL.
2. In the project folder, run **npm start**. The system opens at http://localhost:3000.
3. For phones and the AI voice agent, start the **Cloudflare tunnel** and use the address it shows.
4. Staff log in at **/admin** with the admin password stored in the private .env file.
5. Medical emergency? Patients are always told to call **108** or go to the nearest emergency department.

No passwords, secret keys or private phone numbers are included in this document.